1. Sutherland 7.4

2. Sutherland 8.2

3. Sutherland 8.3

4. Sutherland 8.4

5. Suppose $f : X \to Y$ is a bijection between topological spaces. Show that the following are equivalent:

- for all sets $A \subset X$, A is open if and only if $f(A)$ is open,
- the functions f and f^{-1} are continuous.

6. Let $X = \mathbb{R}$ and let Y be the subset of $\mathbb{R}^2$ of points of the form (x, x^2) with $x \in \mathbb{R}$. Both X and Y are topological spaces by virtue of being metric spaces; the d_1 distance on $\mathbb{R}^2$ also determines a distance on Y. Show that X and Y are homeomorphic as follows.

a) Show that if $x_n \to x$ in $\mathbb{R}$ then $(x_n, x_n^2) \to (x, x^2)$ in Y.
b) What can you conclude about the map $g : X \to Y$ defined by $g(x) = (x, x^2)$?
c) Show that if (a_n, b_n) is a sequence in Y converging to (a, b) then the sequence also converges in $\mathbb{R}^2$.
d) Using the fact that the projection $\pi_1 : \mathbb{R}^2 \to \mathbb{R}$ is continuous, use part (c) to show that $\pi_1|_Y$ is also continuous.
e) Now find an argument to show that X and Y are homeomorphic.

7. Sutherland 8.5 **(AI Free)** (Wait until Friday to start this one)

8. Sutherland 8.6 (Wait until Friday to start this one)

9. AI Use. Please describe specifically how you used AI to help complete this assignment.